
Where the Admissible Radius Binds: a Correction and a Measured Boundary in Online Conformal Prediction

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Abstract

1 A printed claim that the scorecaster breaks online conformal coverage meets a
2 refutation sharing its first and last authors; this paper joins them and locates the
3 edge of the set that refutation relies on. Conformal PID states long-run coverage
4 for any bounded scorecaster; a corollary keeps it for predictable perturbations of
5 the deployed threshold inside an admissible radius. The claim was acted on; the
6 readout’s placement and the corollary’s derivation are credited by name. At the null
7 scorecaster and the minimal admissible saturator, that radius is *tight*. A legal dead
8 band on the *completed* threshold exits that set at its radius; past it miscoverage is
9 1.000000 against $\alpha = 0.1$: coverage fails outright, not merely its rate. The edge
10 belongs to the saturator–scorecaster pair, not the readout; elsewhere that radius
11 is sufficient only. Under the equally legal $\hat{q} \equiv +b/2$ the failing band covers, as
12 measured; at $\hat{q} \equiv -b/2$ both it and partial adjustment at $w = 0.999$ miscover.

13 1 Introduction

14 A refereed paper states that a scorecaster breaks Conformal PID’s coverage guarantee. A later one,
15 by the same first and last authors, proves in the same notation that a bounded perturbation of the
16 deployed threshold does not. The later cites the earlier, so this is ordinary self-correction. The joining
17 is what this paper adds.

18 An online conformal method [Angelopoulos and Bates, 2023] moves a threshold each round. Forecast
19 stability names, scores, and prices that movement [Tunc et al., 2013, Godahewa et al., 2025, Van Belle
20 et al., 2023, Pritularga and Kourntzes, 2024, Genov et al., 2026, Van Belle et al., 2026]. Varying
21 temporal structure at fixed level came earlier, in forecast verification, matching predictive marginals
22 on real producers [Gneiting et al., 2007, Pinson and Girard, 2012, Worsnop et al., 2018]. Coverage
23 and mean length under-describe a deployed interval [Min et al., 2026, Vaze, 2026]. None of it is
24 claimed here.

25 **The claim and the refutation in print.** Conformal PID [Angelopoulos et al., 2023] adds a *score-*
26 *caster* $\hat{q}_{t+1}$ to a saturating error integrator, asking only that it be predictable and lie in $[-b/2, b/2]$:
27 “any function of the past” (Theorem 1). Anything placed in that slot is already quantified over. The
28 claim reads “[s]ince using a scorecaster (D-part) in C-PID breaks the theoretical coverage guarantee”
29 [Li and Rodríguez, 2025]. It is stated and acted on in that paper’s baselines paragraph, verbatim on
30 refereed printed page 18443. It is refuted twice, with proof. Li et al. [2025, Corollary 2] add to the
31 deployed threshold any predictable d_t with $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ and $\mu_t \in [0, 1]$ predictable. Since
32 $\hat{q}_t + d_t$ still lies in $[-b/2, b/2]$ it is itself a legal scorecaster, so the result “has exactly the CPID
33 error-integration form” and $|E_T| = o(T)$. The same generic-slot argument carries AcMCP’s long-run
34 coverage [Wang and Hyndman, 2024]. Both identifiers and “scorecaster” were searched across arXiv
35 full text, arXiv abstracts, OpenReview, and Semantic Scholar’s six citers of Li and Rodríguez [2025]:
36 none returns both the claim and the corollary. Hu et al. [2026] still call scorecaster choice “arbitrary”
37 and lacking “principled guidance” while asserting valid coverage: their complaint is model selection,
38 not validity. **We claim neither the placement nor the derivation.**

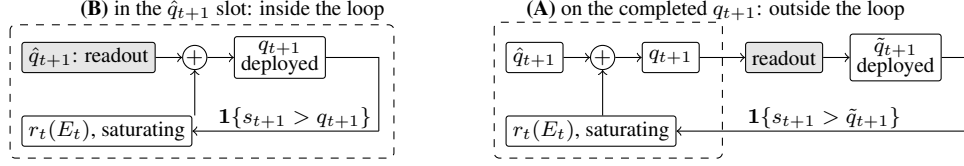


Figure 1: One scalar readout (grey) in its two places in (1); dashed is the loop r_t closes. In (B) the loop closes on the integrator’s own output, and both readouts keep $\{\hat{q}_t\}$ in the convex hull of $\hat{q}_1$ and the raw scorecaster’s range, inside Corollary 2’s admissible set. In (A) it closes on $\tilde{q}_{t+1}$, which it never computed, and nothing keeps the deployed sequence inside that set.

39 **Where Corollary 2 stops, its radius condition binds.** It is a *retention* result, silent outside that
 40 radius, and it leaves open whether the radius is sharp. A legal perturbation leaves it: the L_1 dead-band
 41 family characterised here on the *completed* threshold, whose edge coincides with the corollary’s
 42 radius at the null scorecaster (Section 3). Past it coverage fails outright: the published sufficient
 43 condition is necessary there. Appendix C sets both claims beside four vocabularies for the same
 44 movement.

45 2 Setup

46 **The producer.** A forecaster fit before t induces a score s_t , and a threshold q_t is emitted before
 47 y_t is revealed. The deployed set is $C_t = \{y : s_t(x_t, y) \leq q_t\}$ and $\text{err}_t = \mathbf{1}\{s_t(x_t, y_t) > q_t\}$. The
 48 target is long-run coverage, $T^{-1} \sum_{t \leq T} \text{err}_t = \alpha + o(1)$, with no model on the data. Conformal PID
 49 [Angelopoulos et al., 2023] manipulates the threshold on the score scale:

$$q_{t+1} = \hat{q}_{t+1} + r_t \left(\sum_{i \leq t} (\text{err}_i - \alpha) \right), \quad (1)$$

50 Angelopoulos et al.’s iteration (5). Its r_t saturates: their condition (4) requires $|r_t(x)| \geq b$ with the
 51 sign of x once $|x| \geq c h(t)$, for h nonnegative, nondecreasing, and sublinear. With the accumulator
 52 $E_t = \sum_{i \leq t} (\text{err}_i - \alpha)$, Proposition 2 gives $|E_T| \leq c h(T) + 1$, read on realised scores.

53 **Two readouts, two placements.** Damping a sequentially updated scalar has two standard readouts,
 54 both prior work. A quadratic cost gives linear partial adjustment $\tilde{q}_{t+1} = (1-w)\hat{q}_{t+1}^{\text{raw}} + w\tilde{q}_t$ [Gârleanu
 55 and Pedersen, 2013], published as post-processing [Godaheva et al., 2025] and applied to a deployed
 56 threshold [Binny and Dixit, 2025, Eq. 13]. A proportional cost gives a no-trade band, which moves
 57 only when the target is more than τ away and then pulls back exactly τ [Constantinides, 1986, Davis
 58 and Norman, 1990]. Either readout has two positions in (1), drawn in Figure 1; (A), Placement A, is
 59 measured here. *The dead band (L_1) is not a stylistic alternative to the partial-adjustment (L_2) form;*
 60 *it is the family whose failure Section 3 characterises. Neither is safe by construction: L_2 forfeits*
 61 *Proposition 2’s rate while covering, L_1 past the radius loses coverage, and at the legal $\hat{q} \equiv -b/2$ both*
 62 *lose it.*

63 **The quantity, and the names it already has.** We report the deployed threshold’s *travel*, $\mathcal{T}_H =$
 64 $\sum_{t=2}^H |q_t - q_{t-1}|$, named after actuator travel in process control. The arithmetic is not new, and the
 65 name is a convenience rather than a contribution, used here for one reading only: Appendix C names
 66 ten for the same movement across four vocabularies.

67 **Harness.** The harness fixes $\alpha = 0.1$ and $b = 2$. Its saturator $r_t(x) = b \text{clip}(x/(c h(t)), -1, 1)$,
 68 with $c = 1$ and $h(t) = \log(t+2)$, meets (4) with equality. The scorecaster is null and legal, reducing
 69 (1) to the pure error integration Proposition 2 addresses, so the readout is the only manipulated
 70 variable. Neither choice is neutral: condition (4) is a lower bound met with equality here, and
 71 Theorem 1 permits scorecasters $b/2$ either side of null, predictable but not required to be constant
 72 in t . The adversary is specified, not tie-broken. It plays $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$ with $\varepsilon = 10^{-9}$,
 73 against the *deployed* $\tilde{q}_t$: the $\varepsilon \downarrow 0$ reading of “sets the score at the threshold”. That play fires the
 74 strict indicator and is legal on both sides of Theorem 1’s $[-b/2, b/2]$, so $b/2 + b/2 = b$ is tight.
 75 *That specification names one member of a class:* any adapted $s_t \in [-b/2, b/2]$ is legal, the constant
 76 $s_t \equiv -b/2$ included. Every number in Table 1 (App. A) is measured against the one adversary above;
 77 Section 3’s two boundaries, which differ from each other, are quantified over the whole class.

78 3 The edge of the admissible set

79 **The published retention result.** The radius restated in Section 1 is the one
 80 Li et al. [2025, Corollary 2] prove, and it *retains* $|E_T| \leq ch(T) + 1$. A downstream partial
 81 adjustment of weight w leaves it. Table 1 (App. A) prices that departure: five of the table’s eleven
 82 arms exceed the bound at $T = 10^6$. The rate forfeit is a property of the smoothing *strength*, not
 83 Placement A, and it is bought: travel falls from 28,311 to 202. A readout on the completed q_{t+1}
 84 touches neither r_t nor the integrator’s argument, so it cannot put their condition (4) at risk. It delays a
 85 correction already computed, so smoothing the output feeds E_t rather than damping it.

86 **The L_1 dead band and the four boundaries.** The dead band of Section 2 keeps $|\tilde{q}_{t+1} - q_{t+1}| \leq \tau$
 87 at every round, so “inside Corollary 2” reads $\tau \leq \mu_t(b/2 - |\hat{q}_t|)$. Under saturation the deployed
 88 value is pinned τ below the integrator’s ceiling. Outside the radius two levels decide the answer
 89 and condition (4) names only one of them. Write $A_t^+ = \sup_x r_t(x)$ and $A_t^- = \inf_x r_t(x)$ for the
 90 integrator’s extremes, and $\Lambda_t^+ = \inf_{x \geq ch(t)} r_t(x)$, $\Lambda_t^- = \sup_{x \leq -ch(t)} r_t(x)$ for the levels it is
 91 *guaranteed* to reach once the accumulator clears the radius. Condition (4) says exactly $\Lambda_t^+ \geq b$ and
 92 $\Lambda_t^- \leq -b$. That $\Lambda_t^\pm = A_t^\pm$, i.e. that the integrator reaches its extremes where (4) puts them, is a
 93 further hypothesis, and not one (4) supplies. Set

$$\tau^{*\pm} = \sup_t (|A_t^\pm| \pm \hat{q}_{t+1}) - b/2, \quad \sigma^\pm = \inf_t (|\Lambda_t^\pm| \pm \hat{q}_{t+1}) - b/2. \quad (2)$$

94 **Failure, over the whole admissible class.** If $\tau > \tau^{*+}$ then $\hat{q}_{t+1} + A_t^+ - \tau < b/2$ at every round,
 95 so $\tilde{q}_{t+1} \leq \max(\tilde{q}_t, q_{t+1} - \tau)$ makes $\{\tilde{q} < b/2\}$ absorbing: a release upward lands below $b/2$, and a
 96 hold or a release downward cannot lift the deployed value at all. Against $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$ the
 97 indicator is then $\mathbf{1}\{\tilde{q}_t < b/2\}$ and every round misses. If $\tau \geq \tau^{*-}$ the constant legal play $s_t \equiv -b/2$
 98 makes $\{\tilde{q} \geq -b/2\}$ absorbing by the mirror inequality, so no round can miss and the realised rate
 99 is 0 rather than α . Either way long-run coverage is lost, with $|E_T|$ linear in T . Nothing here asks $\hat{q}$
 100 to be constant, asks the extremes to be attained anywhere, or asks $A_t^\pm$ to be finite beyond what the
 101 hypothesis already forces. **Retention, keyed to the reach rather than the supremum.** If $\tau \leq \sigma^+$
 102 then $E_{t-1} \geq ch(t-1)$ forces $q_t \geq b/2 + \tau$ and so $\tilde{q}_t \geq b/2$, whence $\text{err}_t = 0$ against *every* legal
 103 adversary; if $\tau < \sigma^-$ the mirror forces $\text{err}_t = 1$ below $-ch(t-1)$. The accumulator cannot then
 104 leave $[-ch(t), ch(t)]$ by more than one round, and $|E_T| \leq ch(T) + 1$: Proposition 2’s own bound,
 105 retained verbatim rather than weakened to $o(T)$. Using condition (4) alone this already certifies
 106 $\tau \leq b/2 + \inf_t \hat{q}_t$ and $\tau < b/2 - \sup_t \hat{q}_t$, which for a constant scorecaster is Corollary 2’s radius at
 107 $\mu_t = 1$.

108 **The two directions meet, with no gap, on the sub-class the five settings inhabit.** If $\hat{q}_t \equiv \hat{q}$, if
 109 $\Lambda_t^\pm = A_t^\pm$, and if neither depends on t , then $\sigma^\pm = \tau^{*\pm}$ and **coverage is retained against every legal**
 110 **adapted adversary if and only if $\tau \leq \tau^{*+}$ and $\tau < \tau^{*-}$.** For a symmetric saturator that is one line:
 111 coverage if and only if $\tau \leq \sup_x r_t(x) - b/2 - |\hat{q}|$, with the endpoint included only when $\hat{q} < 0$.
 112 Every setting run here is in that sub-class, so the sharp form applies to all five. **What was printed**
 113 **before is the failure half.** τ^{*+} is the rule Figures 2 and 3 (App. A) draw, and all 19 dead-band runs
 114 agree with it read against the adversary the harness plays. It is not a retention condition, it omits τ^{*-}
 115 entirely, and Table 2 (App. A) separates the two with legal objects only. The supremum can sit where
 116 the dynamics never go: since $|E_t| \leq (1 - \alpha)t$, a saturator equal to the harness’s inside $|x| \leq t^2$ and to
 117 $\pm 10b$ outside it satisfies (4), reads $\tau^{*+} = 19$ and fails at $\tau = 1.001$; and a legal time-varying $\hat{q}$ splits
 118 $\sup_t$ from $\inf_t$, $\hat{q}_t = b/2$ on t a power of two and 0 elsewhere reading $\tau^{*+} = 2$ against $\sigma^+ = 1$.

119 **The sharp form is one number, and its hypotheses are what buy it.** Specialise to a symmetric
 120 saturator that meets (4) with equality and attains its extremes there ($\Lambda_t^\pm = A_t^\pm = \pm b$; the harness’s
 121 $r_t = b \text{ clip}(x/(ch(t)), \pm 1)$ is one, Section 2), and to a constant $\hat{q}$. Then

$$\min(\tau^{*+}, \tau^{*-}) = b/2 - |\hat{q}|, \quad (3)$$

122 with the endpoint admissible only when $\hat{q} < 0$: exactly Corollary 2’s radius at $\mu_t = 1$, now as an
 123 if-and-only-if. **The necessity direction above assumes none of it:** $\tau > \tau^{*+}$, or $\tau \geq \tau^{*-}$, forfeits
 124 coverage against every admissible r_t and every legal predictable $\{\hat{q}_t\}$, extremes attained or not, and
 125 that unrestricted statement is what this section claims. The two hypotheses are load-bearing rather
 126 than decorative: the unattained supremum and the varying- $\hat{q}$ split two paragraphs above are exactly

127 what they rule out. Equation (3) is where the reach meets the supremum, not evidence that they
128 always do.

129 **The mirror boundary on this paper’s grid.** The τ -cliff run’s exact-threshold column, read against
130 the mirror adversary $s_t \equiv -b/2$ (Figure 2, App. A), locates $\tau^{*-} = 1$ at the null scorecaster, closed
131 on the failing side, where (2) puts it. Both 1.000000 and 0.000000 are failures of the same order,
132 $|E_T| = (1 - \alpha)T$ and αT , **so the failure criterion is $|E_T|$, not the miscoverage rate.** Once $\hat{q}$ varies,
133 the miscoverage rate no longer indicates whether coverage has failed.

134 **Both degenerate edges, and what they need.** An unbounded supremum is not on its own enough
135 for $\tau^{*+} = \infty$ to mean what it looks like: the same unreachable-set construction with $r_t(x) = x$
136 past $|x| > t^2$ has no finite supremum and fails at $\tau = 1.5$. What the tangent family has, and that
137 construction does not, is a genuine reach: $\Lambda_t(X) = \tan(\arctan(b)X/(ch(t)))$, so the accumulator
138 level at which the deployed threshold is forced to $b/2$ is $\kappa(\tau)ch(t)$ with $\kappa(\tau) = \arctan(b/2 +$
139 $\tau - \hat{q})/\arctan(b)$, finite at every finite τ . Coverage is retained at every width, as measured; but
140 $\kappa(\tau) > 1$ once $\tau > b/2 + \hat{q}$, so Proposition 2’s own *constant* is forfeited past $\tau = b/2$ while coverage
141 is not. At $\tau = 1.5$, $T = 10^6$ the excursion is 15.10 against Proposition 2’s 14.8155 and against
142 $\kappa(1.5)ch(T) + 1 = 15.8530$. At the other edge, $\sup_t \hat{q}_t = -b/2$ forces $\hat{q} \equiv -b/2$, since every $\hat{q}_t$
143 is at least $-b/2$ already. There $q_{t+1} - \tau \leq -b/2 + b - \tau < b/2$ for every $\tau > 0$ and every t , and
144 $\tilde{q}_1 = 0 < b/2$, so the failing set is absorbing from round 1 with no transient: the admissible window
145 is the single point $\tau = 0$, and $\tau = 0.5$ returns 1.000000, as does partial adjustment at $w = 0.999$.

146 **Tightness of the published condition.** At $\mu_t = 1$, $\hat{q} \equiv 0$, $\tau = 0.9$ covers at 0.100012 while
147 $\tau = 1.5$ returns 1.000000 with $\max_t |E_t| = 900,000$, 60,747 times Proposition 2’s bound at
148 $T = 10^6$ (Table 1, App. A); (2) makes that an if-and-only-if against the whole legal adversary class,
149 not a pattern across five settings. **The claim is that necessity: a tightness result about someone**
150 **else’s sufficient condition**, smaller and more precise than a claim that post-processing forfeits the
151 rate. Elsewhere it stays sufficient only, and provably so: the tangent integrator above forfeits no
152 coverage at any width Corollary 2 permits. At $\hat{q} \equiv +b/2$ the looseness goes the other way and
153 disappears: Corollary 2 permits radius 0, and (2) permits no width at all against the mirror adversary,
154 the unsmoothed control included. Against the specified adversary $\tau = 1.5$ covers there. That
155 asymmetry makes “tight” a located claim: a tie-convention property of the cited closed statements.

156 **The boundary read as a bet.** The boundary has a betting reading (App. B): under a Bernoulli(α)
157 null on err_t , imposed for that reading alone and on nothing else here, an anytime-valid test convicts
158 the $\tau = 1.5$ arm past the edge at the 5% level and cannot reject inside it at any horizon. It adds a
159 diagnostic and a reading, not a result. No result above assumes that null or is certified by that test.

160 4 Limitations

161 **The retention direction is proved with no gap only where the two boundaries meet:** a constant
162 scorecaster and a saturator attaining its extremes at the condition-(4) radius, which is every setting run
163 here; a genuinely time-varying $\hat{q}$ leaves the band $\sigma^+ < \tau \leq \tau^{*+}$ open in neither direction (Section 3).
164 And the excursion is specific to the saturator’s *level*, which condition (4) bounds only from below.
165 **The inherited guarantee is weak where it applies.** Under the tangent integrator $h(t) = t/\log t$, the
166 certified gap $(ch(T) + 1)/T$ decays as $O(1/\log T)$ rather than $O(1/T)$. The forfeited certificate
167 was not a tight one. **The dead-band widths are absolute, and the failure is total only against**
168 **the adversary.** τ is in score units with $b = 2$, so the failing $\tau = 1.5$ is $0.75b$. Under i.i.d. scores it
169 returns 0.249376 at $T = 10^6$, an adversary-free confirmation of the mechanism rather than a retreat.
170 The threshold pins at $b - \tau = 0.5$, and uniform scores on $[-b/2, b/2]$ put $P(s > 0.5) = 0.25$ exactly.
171 **The harness was rebuilt against numbers already read**, and that carries less weight than a rerun
172 done without sight of them. **Placement B is conceded, not tested.** **The i.i.d. check above used**
173 **one seed** (seed 20260820, $T = 10^6$). **The primary, adversarial harness uses none**, and **only its**
174 **null-scorecaster edge is bracketed**; the others are one-sided. **What remains open is the band**
175 **between the two boundaries.** For a genuinely time-varying $\hat{q}$, coverage in $\sigma^+ < \tau \leq \tau^{*+}$ is neither
176 proved nor falsified; Table 2’s power-of-two rows sit there, and closing that band is future work, not a
177 sweep this paper has already run.

References

- Lachlan L. H. Andrew, Siddharth Barman, Katrina Ligett, Minghong Lin, Adam Meyerson, Alan Roytman, and Adam Wierman. A tale of two metrics: Simultaneous bounds on competitiveness and regret. In *Proceedings of the 26th Conference on Learning Theory (COLT 2013)*, pages 741–763, 2013.
- Anastasios N. Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023. doi: 10.1561/22000000101.
- Anastasios N. Angelopoulos, Emmanuel J. Candès, and Ryan J. Tibshirani. Conformal PID control for time series prediction. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023. doi: 10.52202/075280-1000.
- Allen Emmanuel Binny and Anushri Dixit. Who moved my distribution? Conformal prediction for interactive multi-agent systems. arXiv:2511.11567v1, 2025.
- Jiadong Chen, Yang Luo, Xiuqi Huang, Fuxin Jiang, Yangguang Shi, Tieying Zhang, and Xiaofeng Gao. IPOC: An adaptive interval prediction model based on online chasing and conformal inference for large-scale systems. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 202–212, 2023. doi: 10.1145/3580305.3599396.
- George M. Constantinides. Capital market equilibrium with transaction costs. *Journal of Political Economy*, 94(4):842–862, 1986. doi: 10.1086/261410.
- M. H. A. Davis and A. R. Norman. Portfolio selection with transaction costs. *Mathematics of Operations Research*, 15(4):676–713, 1990. doi: 10.1287/moor.15.4.676.
- Ricarda Duerst, Jonas Schöley, Julia Hellstrand, and Mikko Myrskylä. Calibrating probabilistic forecast paths on past forecast errors: an application to the Finnish total fertility rate. Technical Report WP-2024-016, Max Planck Institute for Demographic Research, 7 2024.
- Théo Dupuy, Binbin Xu, Stéphane Perrey, Jacky Montmain, and Abdelhak Imoussaten. Relevance-aware thresholding in online conformal prediction for time series. In *Artificial Intelligence for Healthcare, and Hybrid Models for Coupling Deductive and Inductive Reasoning: First International Joint Conference, HC@AIxIA+HYDRA 2025, Bologna, Italy, October 25–26, 2025, Proceedings*, volume 2830 of *Communications in Computer and Information Science*, pages 196–217, 2026. doi: 10.1007/978-3-032-16708-8_17.
- Uwe Ehret. Convergence index: a new performance measure for the temporal stability of operational rainfall forecasts. *Meteorologische Zeitschrift*, 19(5):441–451, 2010. doi: 10.1127/0941-2948/2010/0480.
- Sergio Galeani, Sophie Tarbouriech, Matthew Turner, and Luca Zaccarian. A tutorial on modern anti-windup design. *European Journal of Control*, 15(3-4):418–440, 2009. doi: 10.3166/ejc.15.418-440.
- Ben Gao, Jordan Patracone, Stéphane Chrétien, and Olivier Alata. Conformal online learning of deep Koopman linear embeddings. arXiv:2511.12760v2, 2025.
- Nicolae Gârleanu and Lasse Heje Pedersen. Dynamic trading with predictable returns and transaction costs. *The Journal of Finance*, 68(6):2309–2340, 2013. doi: 10.1111/jofi.12080.
- Evgenii Genov, Julian Ruddick, Christoph Bergmeir, Majid Vafaeipour, Thierry Coosemans, Salvador García, and Maarten Messagie. Balancing forecast accuracy and switching costs in online optimization of energy management systems. *Expert Systems with Applications*, 298:129305, 2026. doi: 10.1016/j.eswa.2025.129305.
- Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, 2021.
- Isaac Gibbs and Emmanuel J. Candès. Conformal inference for online prediction with arbitrary distribution shifts. *Journal of Machine Learning Research*, 25(162):1–36, 2024.

- 224 Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E. Raftery. Probabilistic forecasts, calibration
225 and sharpness. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 69(2):
226 243–268, 2007. doi: 10.1111/j.1467-9868.2007.00587.x.
- 227 Rakshitha Godahewa, Christoph Bergmeir, Zeynep Erkin Baz, Chengjun Zhu, Zhangdi Song, Salvador
228 García, and Dario Benavides. On forecast stability. *International Journal of Forecasting*, 41(4):
229 1539–1558, 2025. doi: 10.1016/j.ijforecast.2025.01.006.
- 230 Dongjian Hu, Junxi Wu, Shu-Tao Xia, and Changliang Zou. Distribution-informed online conformal
231 prediction. arXiv:2512.07770v2, 2026.
- 232 Ankit Lade, Sai Krishna J., and Indar Kumar. Bias-corrected adaptive conformal inference for
233 multi-horizon time series forecasting. arXiv:2604.13253v1, 2026.
- 234 Ruipu Li and Alexander Rodríguez. Neural conformal control for time series forecasting. In
235 *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 18439–18447,
236 2025. doi: 10.1609/aaai.v39i17.34029.
- 237 Ruipu Li, Daniel Menacho, and Alexander Rodríguez. Optimization-based online conformal predic-
238 tion for multi-step forecasting. arXiv:2508.13362v3, 2025.
- 239 Yizhou Min, Yizhou Lu, Lanqi Li, Zhen Zhang, and Jiaye Teng. Questioning the coverage-length
240 metric in conformal prediction: When shorter intervals are not better. arXiv:2601.21455v2, 2026.
- 241 F. Pappenberger, H. L. Cloke, A. Persson, and D. Demeritt. HESS Opinions “on forecast
242 (in)consistency in a hydro-meteorological chain: curse or blessing?”. *Hydrology and Earth
243 System Sciences*, 15(7):2391–2400, 2011. doi: 10.5194/hess-15-2391-2011.
- 244 P. Pinson and R. Girard. Evaluating the quality of scenarios of short-term wind power generation.
245 *Applied Energy*, 96:12–20, 2012. doi: 10.1016/j.apenergy.2011.11.004.
- 246 Kandrika Pritularga and Nikolaos Kourentzes. Forecast congruence: A quantity to align forecasts
247 and inventory decisions. SSRN working paper 4711817, 2024.
- 248 Aaditya Ramdas, Peter Grünwald, Vladimir Vovk, and Glenn Shafer. Game-theoretic statistics and
249 safe anytime-valid inference. *Statistical Science*, 38(4):576–601, 2023. doi: 10.1214/23-STS894.
- 250 Pouria Sarhadi. On the standard performance criteria for applied control design: PID, MPC or
251 machine learning controller? arXiv:2503.14379v3, 2025.
- 252 Glenn Shafer and Vladimir Vovk. *Game-Theoretic Foundations for Probability and Finance*. Wi-
253 ley Series in Probability and Statistics. Wiley, 2019. ISBN 9780470903056. doi: 10.1002/
254 9781118548035.
- 255 Huseyin Tunc, Onur A. Kilic, S. Armagan Tarim, and Burak Eksioğlu. A simple approach for
256 assessing the cost of system nervousness. *International Journal of Production Economics*, 141(2):
257 619–625, 2013. doi: 10.1016/j.ijpe.2012.09.022.
- 258 Jente Van Belle, Ruben Crevits, and Wouter Verbeke. Improving forecast stability using deep learning.
259 *International Journal of Forecasting*, 39(3):1333–1350, 2023. doi: 10.1016/j.ijforecast.2022.06.
260 007.
- 261 Jente Van Belle, Honglin Wen, Wouter Verbeke, and Pierre Pinson. Stabilizing distribution-free
262 probabilistic forecasts. arXiv:2605.28531v1, 2026.
- 263 Rahul Vaze. Simultaneous coverage and efficiency guarantee in online conformal prediction.
264 arXiv:2607.26577v1, 2026.
- 265 Vladimir Vovk and Ruodu Wang. E-values: Calibration, combination and applications. *The Annals
266 of Statistics*, 49(3):1736–1754, 2021. doi: 10.1214/20-AOS2020.
- 267 Xiaoqian Wang and Rob J. Hyndman. Online conformal inference for multi-step time series forecast-
268 ing. arXiv:2410.13115v2, 2024.

- 269 Rochelle P. Worsnop, Michael Scheuerer, Thomas M. Hamill, and Julie K. Lundquist. Generating
270 wind power scenarios for probabilistic ramp event prediction using multivariate statistical post-
271 processing. *Wind Energy Science*, 3(1):371–393, 2018. doi: 10.5194/wes-3-371-2018.
- 272 Junxi Wu, Dongjian Hu, Yajie Bao, Shu-Tao Xia, and Changliang Zou. Error-quantified conformal
273 inference for time series. arXiv:2502.00818v2, 2025.
- 274 Marco Zanotti. Analyzing the retraining frequency of global forecasting models: towards more stable
275 forecasting systems. arXiv:2506.05776v3, 2026.
- 276 Mingyi Zheng, Hongyu Jiang, Yizhou Lu, and Jiaye Teng. Smoothing-based conformal prediction
277 for balancing efficiency and interpretability. arXiv:2509.22529v1, 2025.
- 278 Ervin Zsótér, Roberto Buizza, and David Richardson. “Jumpiness” of the ECMWF and Met Office
279 EPS control and ensemble-mean forecasts. *Monthly Weather Review*, 137(11):3823–3836, 2009.
280 doi: 10.1175/2009MWR2960.1.

281 Code and data

282 The harness, its frozen configuration, the `results/` files behind every figure and table, the generators
283 that read them, and this project’s audit trail are at [https://\[REPOSITORY LINK - INSERTED BY](https://[REPOSITORY LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION])
284 [AUTHOR PRIOR TO SUBMISSION\]](https://[REPOSITORY LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION]).

285 A Supporting measurements

Table 1: Placement A: a one-scalar readout on the *completed* threshold, under one deterministic adversary, on one pass to $T = 10^6$; all eleven arms run are listed. Proposition 2’s bound is 13.2061 at $T = 2 \times 10^5$ and 14.8155 at $T = 10^6$; “ratio” is $\max_t |E_t|$ over it, and “Sat.” the fraction of rounds on which (4) delivers its full $\pm b$. Every number is measured against the adversary Section 2 specifies.

Readout on the completed threshold	miscov. 2×10^5	miscov. 10^6	$\max_t E_t $ 10^6	ratio	Sat. 10^6	travel 10^6
none (control)	0.100035	0.100007	7.80	0.53	0.0000	28,311
partial adj. $w = 0.5$	0.100030	0.100007	7.70	0.52	0.0000	18,113
partial adj. $w = 0.9$	0.100030	0.100007	7.50	0.51	0.0000	4,081
partial adj. $w = 0.99$	0.100040	0.100006	63.00	4.25	0.0007	1,023
partial adj. $w = 0.999$	0.100035	0.100004	623.70	42.10	0.0097	202
growing time constant $p = 0.5$	0.100030	0.100006	12.60	0.85	0.0000	330
growing time constant $p = 0.9$	0.100030	0.100024	53.00	3.58	0.3181	16
running mean	0.099940	0.100010	49.50	3.34	0.3647	9
dead band $\tau = 0.5$	0.100035	0.100005	10.60	0.72	0.0000	2,045
dead band $\tau = 0.9$	0.100060	0.100012	13.20	0.89	0.0038	1,051
dead band $\tau = 1.5$	1.000000	1.000000	900,000	60,747	1.0000	0.5

Table 2: Legal $(r_t, \hat{q})$ configurations on which the printed τ^* over-certifies, each under two legal adversaries. “Proved window” is where (2) certifies retention against every legal adversary; between it and the failure boundaries $\tau^{*\pm}$ neither direction is claimed. The rows built on a supremum that the dynamics cannot reach, or that is never attained, sit in that gap. Here $\emptyset$ marks a configuration for which no width is certified, $\tau = 0$ included, and $T = 10^5$ except the two power-of-two rows, at $T = 10^6$. The last row carries no dead band at all.

$(r_t, \hat{q})$ configuration	τ	printed τ^*	proved window	$s_t = \hat{q}_t + \varepsilon$		$s_t \equiv -b/2$	
				miscov.	$\max_t E_t $	miscov.	$\max_t E_t $
$\hat{q} \equiv +0.4$, level b	0.599	1.4	$\tau < 0.6$	0.100010	7.20	0.099890	11.60
$\hat{q} \equiv +0.4$, level b	0.8	1.4	$\tau < 0.6$	0.100050	8.70	0.000000	10,000
$\sup_x r_t = 10b$, off the reach	1.001	19	$\tau < 1$	1.000000	90,000	0.000000	10,000
$\sup_x r_t$ unbounded, off the reach	1.5	∞	$\tau < 1$	1.000000	90,000	0.000000	10,000
$A^+ = 4b$, $A^- = -b$	1.5	7	$\tau < 1$	0.100020	4.30	0.000000	10,000
level $4b$, $\hat{q} \equiv 0$	6.999	7	$\tau < 7$	0.100070	12.20	0.100050	11.60
level $4b$, $\hat{q} \equiv 0$	7.0	7	$\tau < 7$	0.100090	12.20	0.000000	10,000
$\sup_x r_t = 2b$, unattained	2.99	3	$\tau < 1$	0.100320	211.10	0.098010	210.50
$\sup_x r_t = 2b$, unattained	3.0	3	$\tau < 1$	1.000000	90,000	0.000000	10,000
$\sup_x r_t = 2b$, attained	3.0	3	$\tau < 3$	0.100000	12.20	0.000000	10,000
$\hat{q}_t = b/2$ on powers of 2	1.0	2	$\emptyset$	0.100009	14.30	0.000000	100,000
$\hat{q}_t = b/2$ on powers of 2	1.9	2	$\emptyset$	0.110960	84,745.70	0.000000	100,000
$\hat{q} \equiv +b/2$, no band	0.0	2	$\emptyset$	0.100000	0.90	0.000000	10,000

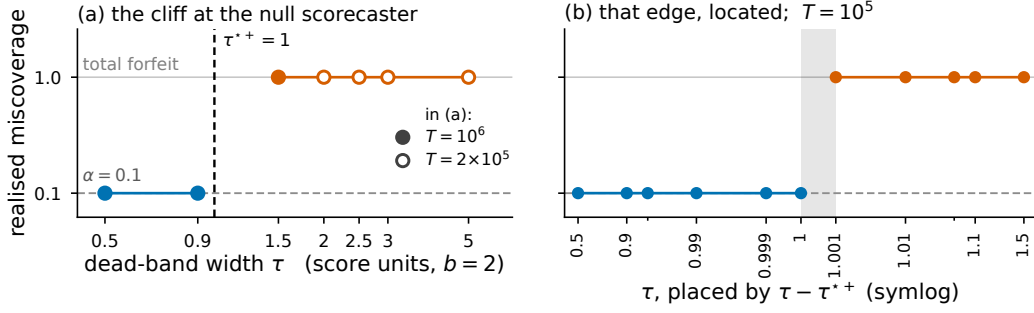


Figure 2: Realised miscoverage against dead-band width τ at the null scorecaster, $\hat{q} \equiv 0$ with a saturator of level b , under the *specified* adversary, where the upward boundary τ^{*+} of (2) is 1. **(a)** The full sweep. Filled markers run to $T = 10^6$ and the four widest bands, drawn open, to $T = 2 \times 10^5$; blue covers and orange forfeits. The dashed rule is at τ^{*+} and the grey band spans the widest covering width to the narrowest failing one. **(b)** The same setting at $T = 10^5$ over eleven measured widths, placed by $\tau - \tau^{*+}$ on a symlog axis that magnifies the neighbourhood of τ^{*+} , so visual spacing there is not proportional to true distance in τ . The linear zone is $|\tau - \tau^{*+}| \leq 10^{-3}$, with unlabelled minor ticks at $\tau = 0.95$ and 1.05 . Figure 3 gives the other four $(r_t, \hat{q})$ settings.

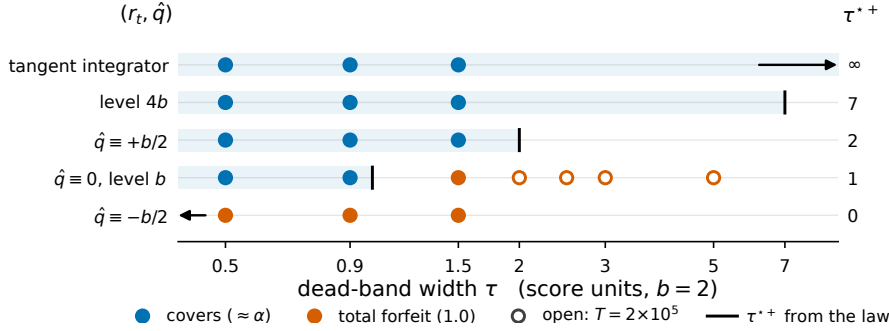


Figure 3: All 19 dead-band runs at the five admissible $(r_t, \hat{q})$ settings, rows ordered by τ^* on the shared horizontal τ axis: blue markers cover at $\approx \alpha$ and orange markers return 1.000000, filled to $T = 10^6$ and open to $T = 2 \times 10^5$, all under the *specified* adversary. The rule on each row is that setting's upward boundary $\tau^{*+} = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$, printed at the right, with the tinted band $\tau \leq \tau^{*+}$; two rules fall off the axis and are drawn as arrows, $\tau^{*+} = 0$ at $\hat{q} \equiv -b/2$ and τ^{*+} unbounded under Angelopoulos et al.'s tangent integrator. The mirror boundary τ^{*-} of (2) is not drawn here and differs from τ^{*+} at the two non-null scorecasters.

286 B The betting reading

287 The boundary is stated in $|E_T|$, which is the net gain of a Skeptic who pays α each round for err_t ;
 288 that is what gives it a betting reading. The reading Section 3 points at is Ville's inequality applied
 289 to the standard martingale $M_t(\lambda) = \prod_{i \leq t} (1 + \lambda(\text{err}_i - \alpha))$, whose logarithm depends on the path
 290 only through t and E_t [Shafer and Vovk, 2019, Ramdas et al., 2023, Vovk and Wang, 2021], under
 291 the imposed null that $\text{err}_t | \mathcal{F}_{t-1}$ is Bernoulli(α), with $\mathcal{F}_{t-1}$ the history through round $t-1$. Ville's
 292 inequality needs each bet to be predictable, fixed by the history before the round it is scored on,
 293 so the direction of failure cannot be read off the realised path first. The diagnostic is therefore the
 294 two-sided mixture $M_t = \frac{1}{2} M_t(+1) + \frac{1}{2} M_t(-1)$, which is a test martingale for the same null because
 295 a convex combination of test martingales is one, and which fixes no direction in advance; both signs
 296 lie in $[-1/(1-\alpha), 1/\alpha]$, the range on which $M_t(\lambda) \geq 0$ on every path, the magnitude 1 is the same
 297 for every arm and both directions, and the mixture costs a factor of two against a correctly guessed
 298 direction. So read, an anytime-valid test of the deployed intervals' calibration convicts the $\tau = 1.5$
 299 arm on its sixth interval at the 5% level, where $M_5 = 12.38$ and $M_6 = 23.52$ against the rejection
 300 threshold $1/0.05 = 20$. Inside the edge the same test cannot reject at that level at any horizon: it
 301 peaks at 2.03 for $\tau = 0.5$ and 3.92 for $\tau = 0.9$, and from round 62 on Proposition 2's bound on
 302 $|E_t|$ forbids rejection on every retained trajectory. M_t is an e-value for that imposed null alone; the

303 accumulator itself is signed and is neither an e-value nor an e-process, and no result in Section 3
 304 assumes that null or is certified by that test.

305 C Related work

306 **One quantity, four vocabularies.** Four literatures name the movement of a sequentially updated
 307 forecast in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC,
 308 MQC/SMQC, congruence, nervousness); *jumpiness* [Zsótér et al., 2009], *convergence index* [Ehret,
 309 2010], *(in)consistency* [Pappenberger et al., 2011]; and *switching cost*. The same arithmetic is the
 310 Multi-Quantile Change over a predictive distribution [Zanotti, 2026, v3, §3.4.1], IACER in applied
 311 control [Sarhadi, 2025], and a path length in dynamic-regret online learning. Online conformal
 312 prediction gives distribution-free long-run coverage for any bounded scorecaster [Angelopoulos et al.,
 313 2023, Thm. 1, Prop. 2]. It generalises the running-error feedback that adaptive conformal inference
 314 [Gibbs and Candès, 2021] and its distribution-shift variant DtACI [Gibbs and Candès, 2024] already
 315 apply directly to the threshold. Forecast stability treats the movement as a functional. It is read out as
 316 post-processing [Godahewa et al., 2025, Eq. 3] and priced against a decision to switch [Genov et al.,
 317 2026, §4.2], [Van Belle et al., 2026].

318 The object that settles a deployed interval’s validity (the admissible set around the deployed threshold)
 319 sits in the first of those four literatures. They are where this paper’s claim and its concessions have
 320 to be placed. **What is conceded, by name.** The placement is prior work: Angelopoulos et al.
 321 [2023, Thm. 1, Prop. 2] state coverage, with a rate, for any admissible integrator and any scorecaster,
 322 and they run a smoothing-family model in that slot themselves. This paper proves that admissible
 323 set tight. Dupuy et al. [2026, Appendix A] give the generic argument, and Duerst et al. [2024]
 324 adopt that scorecaster. The derivation is prior work too. That concession matters most. Li et al.
 325 [2025, Corollary 2] prove the retention statement in Conformal PID’s own notation, with Wang and
 326 Hyndman [2024] an independent second instance. Section 2 concedes the smoother object, the metric
 327 arithmetic, and both readout maps. Zheng et al. [2025], Wu et al. [2025], Chen et al. [2023], Lade
 328 et al. [2026] each place a smoother inside their own validity argument. In one more vocabulary,
 329 this is integrator anti-windup with a feedforward path: free-until-the-bound-binds is its defining
 330 specification, not a finding [Galeani et al., 2009]. Nearest in the other direction, Gao et al. [2025]
 331 run the same recursion but gate the *model* update on the conformal score. Their one non-standard
 332 assumption is that $\sum_{t \leq T} q_t \leq O(\alpha h(T))$ for a sublinear h . They support it with a plot rather than
 333 deriving it. Verification’s matched-level design tests real producers from 2012, with a control stronger
 334 than the pair: the whole marginal [Pinson and Girard, 2012, Worsnop et al., 2018]. Switching-cost
 335 online learning adds a trade-off: sublinear regret is bought by growing θ with T , and growing θ grows
 336 the ratio [Andrew et al., 2013, Thm. 7].

337 NeurIPS Paper Checklist

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339 Question: Do the main claims made in the abstract and introduction accurately reflect the
 340 paper’s contributions and scope?

341 Answer: [\[Yes\]](#)

342 Justification: The abstract and Section 1 state exactly two claims and no more: that this
 343 paper joins an already-refereed claim to its already-published refutation (Conformal PID’s
 344 scorecaster slot vs. the deployed-threshold Corollary 2), and that the admissible radius
 345 Corollary 2 relies on is *tight* at the null scorecaster paired with the minimal admissible
 346 saturator, proved as a necessary-and-sufficient condition on that sub-class in Section 3. Both
 347 the abstract’s hedges (“elsewhere the condition is sufficient only”, the two degenerate-case
 348 caveats) and Section 4’s concessions match the proved scope; no claim in the abstract
 349 exceeds what Section 3’s theorem and Section 4 state.

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 352 made in the paper.

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Answer: [Yes]

Justification: Section 4 is a dedicated Limitations section listing, by name: that the no-gap retention direction is proved only where the two boundaries meet (a constant scorecaster with a saturator attaining its extremes at the condition-(4) radius, which is every setting run); that a genuinely time-varying $\hat{q}$ leaves an open band neither proved nor falsified; that the tangent-integrator’s inherited guarantee is asymptotically weak ($O(1/\log T)$ vs. $O(1/T)$); that the i.i.d. sanity check uses a single seed while the primary adversarial harness uses none; that only the null-scorecaster edge is bracketed on both sides (the other four settings are one-sided); and that the harness was rebuilt against numbers already read rather than run blind. These are strong, specific hedges, not a generic disclaimer.

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Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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Justification: Section 3 states the full admissible class (Conformal PID’s condition (4) and Theorem 1’s bound), defines the two boundary quantities $\tau^{*\pm}$ and $\sigma^\pm$ in (2), and proves both the failure direction (over the whole admissible class, no attainment or constancy assumed) and the retention direction (keyed to the guaranteed reach), showing the two meet with

no gap on the sub-class every tested setting inhabits; the closed form (3) is stated as an if-and-only-if with its two enabling hypotheses named explicitly as load-bearing rather than decorative. **One honest caveat:** the proof is presented in narrative prose within the section text, with assumptions stated in the surrounding sentences and conditions given as labelled equations, rather than as a numbered theorem/proof LaTeX environment with a formal statement–proof separation. The mathematical content (assumptions, both proof directions, both degenerate edge cases) is complete and was independently re-derived and adversarially attacked in this project’s own process; converting the presentation to a numbered theorem environment is a formatting change deferred to a future revision, not a gap in the argument.

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5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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Justification: The simulator, its frozen configuration, the results/ JSON files behind every figure and table, and the generators that read them exist and are complete, but the repository is not yet public: the paper’s “Code and data” section (printed only under the non-anonymous option) currently carries an explicit placeholder, [REPOSITORY LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION], rather than a live URL. The real link is inserted before submission by the author; at the time of this checklist it is not yet available for a reviewer to follow.

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Justification: This paper runs closed-form and vectorized coverage simulations, not model training, so there are no data splits or optimizers; the analogous details are all given in Section 2’s Harness paragraph and Section 3: $\alpha = 0.1$, $b = 2$, the saturator meeting condition (4) with equality, the specified and mirror adversaries, the horizons swept ($T \in \{10^4, 2 \times 10^5, 10^6\}$, with 10^5 for the boundary-stress table), and the dead-band widths τ used at each setting.

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- The assumptions made should be given (e.g., Normally distributed errors).
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